RAJALAKSHMI ENGINEERING COLLEGE

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C2

Started on	Thursday, 17 September 2026, 8:22 AM
State	Finished
Completed on	Thursday, 17 September 2026, 8:23 AM
Time taken	49 secs
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 10.00 out of 10.00

Playing with Numbers:

Ram and Sita are playing with numbers by giving puzzles to each other. Now it was Ram term, so he gave Sita a positive integer 'n' and two numbers 1 and 3. He asked her to find the possible ways by which the number n can be represented using 1 and 3.Write any efficient algorithm to find the possible ways.

Example 1:

Input: *6*
Output:*6*
Explanation: *There are 6 ways to 6 represent number with 1 and 3*

1+1+1+1+1+1
3+3
1+1+1+3
1+1+3+1
1+3+1+1
3+1+1+1

Input Format

First Line contains the number n

Output Format

Print: The number of possible ways 'n' can be represented using 1 and 3

Sample Input

6

Sample Output

6

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>

int main() {
    int n;

    // Read the input silently without any extra text prompts
    if (scanf("%d", &n) != 1) return 0;

    // Handle small inputs directly using base cases
    if (n == 0) {
        printf("1\n");
        return 0;
    } else if (n == 1 || n == 2) {
        printf("1\n");
        return 0;
    }

    // Dynamic Programming variables tracking the last 3 states
    // w0 represents W(i-3), w1 represents W(i-2), w2 represents W(i-1)
    long long w0 = 1; // W(0)
    long long w1 = 1; // W(1)
    long long w2 = 1; // W(2)
    long long current_ways = 0;

    // Compute the ways iteratively up to n
    for (int i = 3; i <= n; i++) {
```

```
    for (int i = 3; i <= n; i++) {
        current_ways = w2 + w0; // W(i) = W(i-1) + W(i-3)

        // Shift states forward for the next iteration
        w0 = w1;
        w1 = w2;
        w2 = current_ways;
    }

    // Print the final result followed by a newline
    printf("%lld\n", current_ways);

    return 0;
}
```

	Input	Expected	Got	
✔	6	6	6	✔
✔	25	8641	8641	✔
✔	100	24382819596721629	24382819596721629	✔

Passed all tests! ✔

Correct

Marks for this submission: 10.00/10.00.

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